

Constraint Inventory for the Exact Certification of the Level Profile $\zeta(a) = (5a - 4)/4$

Companion documentation to Appendix A
of “An Explicit Semiprime-Tail Bound in Square-Root Intervals
and a One-Hypothesis Reduction for $[4n^2 - n, 4n^2 + n]$ ”

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Abstract

This note is the full constraint inventory referred to in Appendix A of the paper referenced above (henceforth [P]). It derives every linear inequality appearing in the certification script `fm_exact.py` from the published theorem that inequality encodes; states the sign, negation, and boundary conventions needed to read the code’s coefficient tuples; records the three places where the encoding is a deliberate *relaxation* of the true tool, and proves that the certification remains sound there by a sandwich against the Coverage Lemma of [P]; and cross-references every atom family to the code identifiers and to the corresponding statements of [P]. This is designed to be an audit trail, not new mathematics.

1 Purpose and provenance

The script `fm_exact.py` certifies, in exact rational arithmetic, the level profile

$$\zeta(\gamma) = \frac{6 - 5\gamma}{4} \quad (\gamma \in [1, 6/5]),$$

in both directions: for $\beta < \zeta(\gamma)$ every estimation block admits an admissible configuration (achievability — proved by hand in [P], Coverage Lemma, and re-verified here as a byproduct), and at $\beta = \zeta(\gamma)$ a block exists that no encoded tool covers (optimality — [P], Optimality Proposition). The certification is a finite computation because every tool’s admissibility region is a conjunction of inequalities that are *linear* in the block coordinates at fixed γ . This document is the dictionary between those inequalities, the code, and the literature.

2 Conventions

Convention 2.1 (coordinates and exponents). A slice is fixed by $\gamma \in [1, 6/5]$ (prime block $p \sim N^\gamma$; equivalently $a = 2 - \gamma$, cofactor $q \asymp N^a$). The unknowns are $x = (\mu, \eta, \beta)$: the Vaughan-piece exponent, the dyadic frequency exponent, and the sieve-level exponent. All exponents are with respect to base $N = 2n$; $x_{\text{arith}} := 4n^2 \asymp N^2$. We write $\Sigma := \beta + \gamma$ and

$\xi := \eta + 2 - \Sigma$ (phase-size exponent). The admissibility requirement for a block, [P] eq. (trilinear requirement), is that the estimating bound have exponent at most Σ (i.e. relative saving N^η against the trivial size $N^{\eta+\Sigma}$), with an $x^{-\varepsilon}$ margin suppressed throughout this document: all atoms are stated with $\varepsilon = 0$ and *closed* inequalities. Facet equalities are therefore boundary points of both the pass and fail regions; this is intentional (see Convention 2.3).

Convention 2.2 (atom format in the code). In `failure_atoms()` a constraint is the tuple $C(a_\mu, a_\eta, a_\beta, c_0, c_\gamma)$, meaning

$$a_\mu \mu + a_\eta \eta + a_\beta \beta \leq c_0 + c_\gamma \gamma,$$

all entries rational. Each *tool* passes on a conjunction of pass conditions; the code stores, per tool, the list of *negated* pass conditions (“fail atoms”), one of which must hold for the tool to fail at a point. A *failure witness* is a point where every tool fails; distributing the disjunctions gives the product over tools of the fail-atom counts:

$$\underbrace{3}_{\text{FI0}} \times \underbrace{3}_{\text{FI1}} \times \underbrace{3}_{\text{FI2}} \times \underbrace{2}_{\text{RS0}} \times \underbrace{2}_{\text{RS1}} \times \underbrace{2}_{\text{RS2}} \times \underbrace{1}_{\text{RS3}} \times \underbrace{1}_{\text{ELEM}} = 216$$

candidate polytopes per piece type (ELEM is present only for type I, where its single fail option is the conjunction of its three negated clauses; type II has 216 combinations without the final factor). Each polytope comprises the chosen fail atoms plus the region constraints of Section 3 (≈ 14 inequalities in 3 unknowns); the minimal β over it is found by exhaustive vertex enumeration ($\binom{14}{3} = 364$ candidate bases, 3×3 systems solved by Cramer’s rule over $\mathbb{Q}$ in `solve3()`), and $\zeta(\gamma)$ is the minimum over all polytopes (`min_beta()`).

Convention 2.3 (boundaries and monotonicity). Because all fail atoms are closed, the computed $\zeta(\gamma)$ is the infimum of levels admitting a witness, and the witness printed at $\beta = \zeta(\gamma)$ sits on the facet with equalities active. Failure is not a priori monotone in β (raising β widens the frequency range and tightens windows, but also widens the reachable grouping intervals), so the script’s output is, by itself, only the statement “the *first* failure occurs at $\beta = \zeta(\gamma)$.” That everything strictly below the facet passes is the Coverage Lemma of [P], proved by hand; the two artifacts meet at the facet. See also Section 7.

3 The block model and the region R

Per [P] (Decomposition and grouping), a block carries variables (h, e_1, e_2, m_v, k) with exponents $(\eta, \beta - s, s, \mu, \gamma - \mu)$, $s \in [0, \beta]$ free. Vaughan pieces: type I has $\mu \in [0, 2\gamma/3]$ with k smooth; type II has $\mu \in [\gamma/3, \gamma/2]$, both coefficients arbitrary. The frequency range from the pipeline lemma is $0 \leq \eta \leq \Sigma - 1$ (blocks with $\Sigma \leq 1$ require nothing). The region constraints in `region()`:

code tuple	inequality	role	piece type
<code>C(-1,0,0,0,0)</code>	$\mu \geq 0$	piece range	I
<code>C(1,0,0,0,2/3)</code>	$\mu \leq 2\gamma/3$	piece range	I
<code>C(-1,0,0,0,-1/3)</code>	$\mu \geq \gamma/3$	piece range	II
<code>C(1,0,0,0,1/2)</code>	$\mu \leq \gamma/2$	piece range	II
<code>C(0,-1,0,0,0)</code>	$\eta \geq 0$	frequency range	both
<code>C(0,1,-1,-1,1)</code>	$\eta \leq \beta + \gamma - 1$	frequency range	both
<code>C(0,0,-1,0,0), C(0,0,1,1,0)</code>	$0 \leq \beta \leq 1$	search box	both

4 Tool family RS: the Robert–Sargos theorem

4.1 Statement and specialization

Robert–Sargos [3, Theorem 1]: for exponents with $\alpha(\alpha-1)\beta_0\gamma_0 \neq 0$, dyadic ranges H, N_1, M_1 , phase size $X \geq 1$, and arbitrary coefficients $|a(h, \nu)| \leq 1$, $|b(w)| \leq 1$,

$$\sum_{h \sim H} \sum_{\nu \sim N_1} a(h, \nu) \sum_{w \sim M_1} b(w) e\left(X \frac{h^{\beta_0} \nu^{\gamma_0} w^{\alpha}}{H^{\beta_0} N_1^{\gamma_0} M_1^{\alpha}}\right) \ll_{\varepsilon} (HN_1M_1)^{1+\varepsilon} \left\{ \left(\frac{X}{HN_1M_1^2}\right)^{1/4} + (HN_1)^{-1/4} + M_1^{-1/2} + X^{-1/2} \right\}.$$

We use $(\beta_0, \gamma_0, \alpha) = (1, -1, -1)$, so $\alpha(\alpha-1)\beta_0\gamma_0 = -2 \neq 0$; the standalone slot w has exponent m_1 , the joint slot (h, ν) has exponents $(\eta, \Sigma - m_1)$, and $X = N^{\xi}$ with $\xi \geq 2 - \Sigma \geq 3/4 > 0$.

4.2 Window derivation

The requirement is $(\eta + \Sigma) + \max_i t_i \leq \Sigma$, i.e. $\max_i t_i \leq -\eta$, where in exponents, with $n_1 = \Sigma - m_1$ and $\xi - \eta = 2 - \Sigma$:

$$t_1 = \frac{\xi - \eta - n_1 - 2m_1}{4} = \frac{2 - 2\Sigma - m_1}{4} \Rightarrow m_1 \geq 4\eta + 2 - 2\Sigma \quad (\text{A})$$

$$t_2 = -\frac{\eta + n_1}{4} \Rightarrow m_1 \leq \Sigma - 3\eta \quad (\text{B})$$

$$t_3 = -\frac{m_1}{2} \Rightarrow m_1 \geq 2\eta \quad (\text{C})$$

$$t_4 = -\frac{\xi}{2} \Rightarrow \eta \leq 2 - \Sigma \quad (\text{D})$$

Subsumptions used by the encoding: (A) $\Leftarrow$ (C) since $4\eta + 2 - 2\Sigma \leq 2\eta \iff \eta \leq \Sigma - 1$, which is a region constraint; (D) is implied on the relevant part of the region (see Remark 4.1). The effective window is $m_1 \in [2\eta, \Sigma - 3\eta]$, matching [P] (Admissibility window Lemma), whose window is nonempty iff $5\eta \leq \Sigma$.

4.3 The four grouping families and their code atoms

The reachable standalone exponents are $m_1 \in \{s, \mu + s, \gamma - \mu + s, \gamma + s\}$, $s \in [0, \beta]$ ([P], available-set display; the fourth family is retained here because the RS window's upper cap $\Sigma - 3\eta$ can exceed γ at small η). Taking in each family the choice $m_1 = \max(\text{base}, 2\eta)$ where possible, the pass conditions and their negations:

tool	pass conditions	fail atoms (code)	fail inequality
RS0 ($m_1 = s$)	$2\eta \leq \beta; 5\eta \leq \Sigma$	$\mathbf{C}(0, -2, 1, 0, 0)$	$\beta \leq 2\eta$
		$\mathbf{C}(0, -5, 1, 0, -1)$	$\beta \leq 5\eta - \gamma$ (i.e. $5\eta \geq \Sigma$)
RS1 ($m_1 = \mu + s$)	$\mu + \beta \geq 2\eta; \mu \leq \Sigma - 3\eta$	$\mathbf{C}(1, -2, 1, 0, 0)$	$\mu + \beta \leq 2\eta$
		$\mathbf{C}(-1, -3, 1, 0, -1)$	$\mu \geq \Sigma - 3\eta$
RS2 ($m_1 = \gamma - \mu + s$)	$\gamma - \mu + \beta \geq 2\eta; \gamma - \mu \leq \Sigma - 3\eta$	$\mathbf{C}(-1, -2, 1, 0, -1)$	$\gamma - \mu + \beta \leq 2\eta$
		$\mathbf{C}(1, -3, 1, 0, 0)$	$\mu + \beta \leq 3\eta$ (i.e. $\gamma - \mu \geq \Sigma - 3\eta$)
RS3 ($m_1 = \gamma + s$)	$3\eta \leq \beta$ (i.e. $\gamma \leq \Sigma - 3\eta$)	$\mathbf{C}(0, -3, 1, 0, 0)$	$\beta \leq 3\eta$

Remark 4.1 (documented relaxations in the RS encoding). Three omissions make the encoded tools slightly *stronger* than the true ones on part of the search box: (i) **RS1** and **RS2** omit the window-nonemptiness condition $5\eta \leq \Sigma$ (which **RS0** includes); (ii) **RS3** omits the lower-edge

condition $\gamma + s \geq 2\eta$, vacuous only when $2\eta \leq \gamma$, i.e. when $\Sigma \leq (\gamma + 2)/2$; (iii) condition (D) of Subsection 4.2 is omitted everywhere, valid only when $\eta \leq 2 - \Sigma$, i.e. when $\Sigma \leq 3/2$ given $\eta \leq \Sigma - 1$. All three omitted conditions *hold automatically* on $\{\Sigma \leq 5/4, \eta \leq \Sigma - 1\}$, which contains the entire facet region; outside it they are genuine relaxations. Soundness is unaffected: see Section 7.

5 Tool family ELEM: elementary bounds (type I only)

For a type I block the smooth variable $k \sim K = N^\kappa$, $\kappa = \gamma - \mu$, is summed with the phase $f(k) = hz'/(ck)$, $c = e_1 e_2 m_v \sim N^\chi$, $\chi = \beta + \mu$, all other variables in absolute value; $|f'| \asymp \Lambda = N^\lambda$, $\lambda = \eta + 2 - \chi - 2\kappa$. Three clauses, whose disjunction is the pass condition; the tool fails only if all three fail, which is the single ELEM conjunction in the code.

(E1) *Kuzmin–Landau branch* ($\lambda < 0$): by Kuzmin–Landau (e.g. [2, Theorem 2.1], after dyadic subdivision so that $|f'| \leq \frac{1}{2}$), $\sum_k e(f(k)) \ll \Lambda^{-1}$; total exponent $\eta + \chi - \lambda = 2(\chi + \kappa) - 2 = 2\Sigma - 2 \leq \Sigma$ for $\Sigma \leq 2$: *always admissible in region*. Pass clause: $\lambda < 0 \iff \eta \leq \beta - \mu + 2\gamma - 2$.

Fail atom $\mathbf{C}(-1, -1, 1, 2, -2)$: $\eta \geq \beta - \mu + 2\gamma - 2$.

(E2) *Second-derivative test / exponent pair* ($\frac{1}{2}, \frac{1}{2}$) [2, Theorem 2.2]: $\sum_k \ll (\Lambda K)^{1/2} + (K/\Lambda)^{1/2}$. The first term needs $\eta + \chi + \frac{\lambda + \kappa}{2} \leq \Sigma$, which simplifies to $3\eta + 2\mu \leq \beta + 3\gamma - 2$; the companion term needs $\eta + \chi + \frac{\kappa - \lambda}{2} = \frac{\eta + 3\Sigma - 2}{2} \leq \Sigma$, i.e. $\eta \leq 2 - \Sigma$, automatic on the facet region.

Fail atom $\mathbf{C}(-2, -3, 1, 2, -3)$: $3\eta + 2\mu \geq \beta + 3\gamma - 2$.

(E3) *Exponent pair* ($\frac{1}{6}, \frac{2}{3}$) [2, §3]: $\sum_k \ll \Lambda^{1/6} K^{2/3} (+ \text{admissible companions as in (E1), (E2)})$; the main term needs $\eta + \chi + \frac{\lambda}{6} + \frac{2\kappa}{3} = \frac{7\eta + 5\beta + 3\mu + 2\gamma + 2}{6} \leq \Sigma$, which simplifies to $7\eta + 3\mu \leq \beta + 4\gamma - 2$.

Fail atom $\mathbf{C}(-3, -7, 1, 2, -4)$: $7\eta + 3\mu \geq \beta + 4\gamma - 2$.

Remark 5.1 (P). uses only (E1) and (E2) in the Coverage Lemma; (E3) is carried in the certification solely so that the Optimality Proposition quantifies over the larger toolkit. The certified optimum is identical with or without (E3) — one of the two byproducts recorded in [P], Appendix A.

6 Tool family FI: Fouvry–Iwaniec Theorem 6

This family is *not used* in the proofs of [P]; it is encoded so that the Optimality Proposition’s claim — the profile is unchanged when [1, Theorem 6] is adjoined — is machine-checked rather than asserted. Since the derivation appears nowhere in [P], it is recorded here in full.

6.1 Statement and exponent form

[1, Theorem 6]: for $\alpha \neq 0, 1$, an additive character $\chi(h)$, and arbitrary $|\varphi_m|, |\psi_n| \leq 1$,

$$S_{\chi\varphi\psi}(H, M, N) \ll (HMN)^{1/2} \left[(H + N)^{1/2} (x_6^{1/8} H^{-1/6} M^{1/12} N^{1/6} + x_6^{1/8} H^{-1/8} N^{3/8} + N^{1/2} + N^{1/4} M^{1/8}) x_6^{1/8} + M^{1/2} + x_6^{-1/4} M^{1/2} N \right] \mathcal{L}^4,$$

where x_6 is the phase size. With slot exponents (m_0, n_0) , $m_0 + n_0 = \Sigma$, frequency η , and $x_6 = N^\xi$, the requirement (bound $\leq N^\Sigma$) yields, after the prefactor $(\eta + \Sigma)/2$:

- the $M^{1/2}$ term (T5): $\frac{\eta+\Sigma}{2} + \frac{m_0}{2} \leq \Sigma \iff \eta \leq n_0$ — a *lower* window edge, which also fixes $(H + N)^{1/2} \mapsto N^{n_0/2}$ in the remaining terms;
- five upper caps on n_0 , derived by direct substitution ($\xi = \eta + 2 - \Sigma$, $m_0 = \Sigma - n_0$):

$$\begin{aligned}
\text{T1 } (x_6^{1/4} H^{-1/6} M^{1/12} N^{1/6}) : \quad & 7n_0 \leq 8\Sigma - 7\eta - 6 \Rightarrow n_0 \leq \text{cap}_1 = \frac{8\Sigma - 7\eta - 6}{7} \\
\text{T2 } (x_6^{1/4} H^{-1/8} N^{3/8}) : \quad & 7n_0 \leq 6\Sigma - 5\eta - 4 \Rightarrow n_0 \leq \text{cap}_2 = \frac{6\Sigma - 5\eta - 4}{7} \\
\text{T3 } (x_6^{1/8} N^{1/2}) : \quad & 8n_0 \leq 5\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_3 = \frac{5\Sigma - 5\eta - 2}{8} \\
\text{T4 } (x_6^{1/8} N^{1/4} M^{1/8}) : \quad & 5n_0 \leq 4\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_4 = \frac{4\Sigma - 5\eta - 2}{5} \\
\text{T6 } (x_6^{-1/4} M^{1/2} N) : \quad & 2n_0 \leq 2 - \Sigma - \eta \Rightarrow n_0 \leq \text{cap}_6 = \frac{2 - \Sigma - \eta}{2}
\end{aligned}$$

(Sample derivation, T2: multiply the requirement $\frac{\eta+\Sigma}{2} + \frac{n_0}{2} + \frac{\xi}{4} - \frac{\eta}{8} + \frac{3n_0}{8} \leq \Sigma$ by 8 and substitute ξ : $5\eta + 7n_0 + 2\Sigma + 4 \leq 8\Sigma$. The others are identical in kind.)

6.2 Pruning lemma

Lemma 6.1. *On $\{\eta \leq \Sigma - 1, \Sigma \leq 4/3\}$: $\text{cap}_2 \leq \text{cap}_1$, $\text{cap}_2 \leq \text{cap}_3$, and $\text{cap}_2 \leq \text{cap}_4$. Hence the FI window is $[\eta, \min(\text{cap}_6, \text{cap}_2)]$.*

Proof. $\text{cap}_1 - \text{cap}_2 = \frac{2\Sigma - 2\eta - 2}{7} \geq 0$ iff $\eta \leq \Sigma - 1$. $\text{cap}_2 \leq \text{cap}_4$ iff $5(6\Sigma - 5\eta - 4) \leq 7(4\Sigma - 5\eta - 2)$ iff $\Sigma + 5\eta \leq 3$; with $\eta \leq \Sigma - 1$ this follows from $6\Sigma \leq 8$, i.e. $\Sigma \leq 4/3$. $\text{cap}_2 \leq \text{cap}_3$ iff $8(6\Sigma - 5\eta - 4) \leq 7(5\Sigma - 5\eta - 2)$ iff $13\Sigma \leq 5\eta + 18$, implied by $\Sigma \leq 4/3 < 18/13$. $\square$

6.3 Grouping families and code atoms

The reachable n_0 families are $\{s, \mu + s, \gamma - \mu + s\}$, $s \in [0, \beta]$; the fourth family $n_0 = \gamma + s$ is *provably dead* for this tool, since $\gamma \geq 1$ while $\text{cap}_6 \leq \frac{2-\Sigma}{2} \leq \frac{1}{2}$, and is therefore not encoded. In each family we take n_0 at the base value (lower edge condition base + $\beta \geq \eta$):

tool	pass conditions	fail atoms (code)	fail inequality
FI0	$\eta \leq \beta$; $\eta \leq \text{cap}_6$; $\eta \leq \text{cap}_2$	$C(0, -1, 1, 0, 0)$	$\eta \geq \beta$
		$C(0, -3, -1, -2, 1)$	$3\eta + \beta \geq 2 - \gamma$ ($\eta \geq \text{cap}_6$)
		$C(0, -12, 6, 4, -6)$	$12\eta - 6\beta \geq 6\gamma - 4$ ($\eta \geq \text{cap}_2$)
FI1	$\mu + \beta \geq \eta$; $\mu \leq \text{cap}_6$; $\mu \leq \text{cap}_2$	$C(1, -1, 1, 0, 0)$	$\mu + \beta \leq \eta$
		$C(-2, -1, -1, -2, 1)$	$2\mu + \eta + \beta \geq 2 - \gamma$
		$C(-7, -5, 6, 4, -6)$	$7\mu + 5\eta - 6\beta \geq 6\gamma - 4$
FI2	$\gamma - \mu + \beta \geq \eta$; $\gamma - \mu \leq \text{cap}_6$; $\gamma - \mu \leq \text{cap}_2$	$C(-1, -1, 1, 0, -1)$	$\gamma - \mu + \beta \leq \eta$
		$C(2, -1, -1, -2, 3)$	$2\mu - \eta - \beta \leq 3\gamma - 2$
		$C(7, -5, 6, 4, 1)$	$7\mu - 5\eta + 6\beta \leq \gamma + 4$

(The last two FI2 rows are the sign-flipped forms of $\gamma - \mu \geq \text{cap}_6$ and $\gamma - \mu \geq \text{cap}_2$.) The pruning is a further relaxation outside $\Sigma \leq 4/3$, handled by the same soundness argument below. The trivial-size lower edge $\eta \leq n_0$ is encoded as the first pass condition in each family.

7 Soundness of the relaxed encoding

Let $\zeta_{\text{code}}(\gamma)$ be the script's output and $\zeta_{\text{true}}(\gamma)$ the first-failure level of the *exactly* encoded toolkit. Every deviation documented in Remark 4.1 and Subsection 6.2 makes an encoded tool pass on a superset of its true pass region; failure witnesses are therefore harder to find, so $\zeta_{\text{code}} \geq \zeta_{\text{true}}$. In the other direction, the Coverage Lemma of [P] proves by hand — using only the exactly-valid parts of the encoding, all active inside $\{\Sigma \leq 5/4\}$ where every omitted condition holds — that all blocks pass for $\beta < (6 - 5\gamma)/4$, so $\zeta_{\text{true}} \geq (6 - 5\gamma)/4$. Since the script returns $\zeta_{\text{code}} = (6 - 5\gamma)/4$ exactly at every tested rational γ , the sandwich closes:

$$\frac{6 - 5\gamma}{4} = \zeta_{\text{code}} \geq \zeta_{\text{true}} \geq \frac{6 - 5\gamma}{4}.$$

Finally, the printed witnesses are legitimate witnesses for the *true* system: at $(\mu, \eta) = (\gamma/2, \Sigma - 1)$ with $4\beta + 5\gamma = 6$ one checks directly that every omitted condition ($5\eta \leq \Sigma$, $\eta \leq 2 - \Sigma$, $2\eta \leq \gamma$, $\Sigma \leq 4/3$) is satisfied, so no relaxed tool was wrongly credited there.

8 Cross-reference table

code identifier	this document	item in [P]
RS0-RS3	§4	§5.4 Admissibility-window Lemma (5.3); §5.3 available-set display (10)
ELEM clause (E1)	§5	§5.5 elementary Lemma (5.4), Kuzmin–Landau term Λ^{-1}
ELEM clause (E2)	§5	§5.5 elementary Lemma (5.4), exponent pair $(\frac{1}{2}, \frac{1}{2})$; used in §5.6 Coverage Lemma (5.5)
ELEM clause (E3)	§5	§5.6 Optimality Prop. (5.6); Appendix A (certification only)
FI0-FI2	§6	§5.6 Optimality Prop. (5.6); Appendix A byproducts (certification only)
region()	§3	§5.3 Decomposition and grouping; §5.1 pipeline Lemma (5.1), frequency range below (9)
solve3(), min.beta()	Convention 2.2	Appendix A (vertex enumeration)

9 Reproduction

Run `python3 fm_exact.py` (Python 3; `numpy/scipy` are used only for the final floating-point root-finding of θ^{**} , which is verification of a closed-form calculus step and not part of the certification). Expected output: the exact table $\zeta(\gamma) = 1/4, 3/16, 1/8, 1/16, 0$ at $\gamma = 1, 21/20, 11/10, 23/20, 6/5$, agreement with $(6 - 5\gamma)/4$ throughout, the witness $(\mu, \eta) = (11/20, 9/40)$ at $\gamma = 11/10$, the threshold $\theta^{**} = 0.939305$, and the margins $+0.0136, +0.2028, +0.5478$ (in units of u) at $\vartheta = 0.94, 0.95, 0.97$. Runtime is a few minutes.

References

- [1] E. Fouvry and H. Iwaniec, *Exponential sums with monomials*, J. Number Theory **33** (1989), 311–333.

- [2] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Math. Soc. Lecture Note Ser. **126**, Cambridge Univ. Press, 1991.
- [3] O. Robert and P. Sargos, *Three-dimensional exponential sums with monomials*, J. reine angew. Math. **591** (2006), 1–20.